Example 3.25

Factorise $7c^2 + 2c - 5$

Product = –35	Sum = 2
1×(–35) = –35	1–35 = –34
–1 × 35 = –35	–1+35 = 34
5 × (–7) = –35	5–7 = –2
–5 × 7 = –35	**–5+7 = 2** ✓

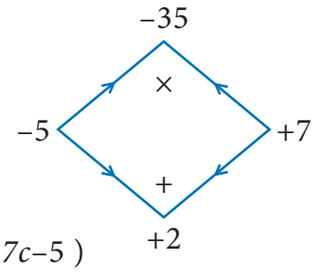
Solution:

Given $7c^2 + 2c - 5$

This is in the form of $ax^2 + bx + c$

We get $a = 7,\ b = 2,\ c = -5$

Now, the product $= a \times c = 7 \times (-5) = -35$ and sum $b = 2$



$= 7c^2 + 2c - 5$

$= 7c^2 - 5c + 7c - 5$ (the middle term $2c$ can be written as $-5c+7c$)

$= (7c^2 - 5c) + (7c - 5)$

$= c(7c - 5) + 1(7c - 5)$ (taking out the common factor $7c-5$)

$= (7c - 5)(c + 1)$

Therefore, $(7c-5)$, $(c+1)$ are the two factors.

Try these

Factorise the following :

1) $3y + 6$ 2) $10x^2 + 15y^2$ 3) $7m(m-5) + 1(5-m)$ 4) $64 - x^2$ 5) x^2-3x+2

6) $y^2-4y-32$ 7) $p^2+2p-15$ 8) $m^2+14m+48$ 9) x^2-x-90 10) $9x^2-6x-8$

3.7.1 Factorisation using cubic identities

The cubic identities are

(i) $(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$

(ii) $(a-b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$

Note

$8a^3 = 2 \times 2 \times 2 \times a^3$

$= 2^3 a^3 \quad = (2a)^3$

I. Factorise using the identity $(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$

Example 3.26

Factorise: $x^3 + 15x^2 + 75x + 125$

Solution:

Given $x^3 + 15x^2 + 75x + 125$

This can be written as $x^3 + 15x^2 + 75x + 5^3$

Comparing with $a^3 + 3a^2b + 3ab^2 + b^3 = (a+b)^3$

we get $a = x,\ b = 5$

The given expression can be expressed as

$(x)^3 + 3(x)^2\,(5) + 3(x)\,(5)^2 + (5)^3 = (x + 5)^3$

$= (x + 5)\,(x + 5)\,(x + 5)$ are the three factors.

Note

Perfect cube numbers

A number which can be written in the form of $x \times x \times x$ is called perfect cube number.

Examples:

$8 = 2 \times 2 \times 2 = 2^3$

$27 = 3 \times 3 \times 3 = 3^3$

$125 = 5 \times 5 \times 5 = 5^3$

Here 8, 27, 125 are some of perfect cube numbers

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